

When Does Search Help LLM Reasoning?

A Topology-Driven Diagnostic of A* versus Chain-of-Thought

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Abstract

Chain-of-Thought (CoT) prompting and search-based reasoning are two dominant inference-time strategies for improving LLM reasoning. CoT commits to a single linear trace, whereas search explores multiple candidate traces. This makes search attractive for difficult reasoning problems, but also raises a basic question: when does the extra exploration help, and when does it introduce unnecessary noise?

We study this question using A* as a controlled priority-search abstraction over natural-language reasoning traces, and compare it with CoT on 2,151 questions from LogicQA, StrategyQA, and HotpotQA using LLaMA 3.1-8B. Aggregate results show no universal winner: A* improves over CoT on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp), but slightly trails on LogicQA (−0.9 pp). However, instance-level analysis reveals substantial complementarity, suggesting that the relevant question is not which method is globally better, but which problem structures favor each inference strategy.

To explain this complementarity, we introduce a **Reasoning Topology Framework** that characterizes each question using 13 pre-generation structural features. Clustering these features yields four topology types and shows that branching complexity, rather than dataset identity alone, predicts when A* helps. A cross-dataset topic analysis independently supports the same conclusion: A* dominates in categories requiring multi-hop evidence chaining, premise scanning, temporal reasoning, or entity-chain resolution, whereas CoT is preferable for single-step inference categories. A topology-aware router that observes no model outputs recovers 26.1% of the oracle gap, outperforming output-based routers such as semantic entropy and LLM-as-critic adjudication.

We further show why this routing problem is difficult: in 38% of divergent cases, the more fluent trace is wrong, and 91.1% of A* failures arise from overthinking. Finally, search amplifies smaller models on multi-hop QA, yielding a $3.5\times$ gain for Qwen-0.5B, but only when the topology rewards exploration. Together, these findings show that search is not a universally stronger replacement for CoT; it is a topology-sensitive computation strategy.

Keywords: A* search, chain-of-thought, LLM reasoning, topic modelling, reasoning topology, adaptive routing

1 Introduction

Chain-of-thought (CoT) prompting (Wei et al., 2022) has become the standard interface for eliciting reasoning from large language models (LLMs). By asking a model to externalise intermediate steps before producing an answer, CoT often improves performance on arithmetic, commonsense, symbolic, and multi-hop reasoning tasks. However, CoT also imposes a strong structural assumption on inference: reasoning is generated as a single linear trajectory,

$$s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_T.$$

Once an early step is produced, all later steps are conditioned on it. Thus, CoT is not merely a prompting trick; it is an inference policy that commits the model to one path through the latent space of possible explanations. This makes CoT

efficient when the solution follows a mostly linear path, but brittle when the problem requires backtracking, comparing alternative hypotheses, aggregating multiple pieces of evidence, or recovering from an initially plausible but incorrect intermediate conclusion.

Search-based reasoning methods relax this linearity assumption. Tree-of-Thoughts (Yao et al., 2024) explores multiple candidate reasoning paths, Graph-of-Thoughts (Besta et al., 2024) allows richer dependencies among intermediate thoughts, and MCTS-based approaches (Hao et al., 2023) use rollout-based exploration to guide inference. These methods are often motivated by the intuition that difficult reasoning requires exploring more than one path. Yet this intuition is incomplete. Search increases the space of candidate reasoning trajectories, but it also increases the risk of over-exploration, redundant decomposition, spurious branches, and unnecessary reasoning on problems that are already solvable by a direct linear chain. The benefit of search therefore depends on whether the problem actually contains useful branching structure.

Most evaluations of search-based reasoning report benchmark-level gains. Such aggregate comparisons are insufficient because benchmark identity is only a coarse proxy for reasoning structure. A HotpotQA question may be nearly linear if the evidence chain is obvious, while a StrategyQA question may require branching through hidden commonsense premises. Conversely, a LogicQA instance may require compact deductive closure rather than broad search. What is needed is an instance-level account of the problem structure that determines whether search is useful.

We ask four nested questions. First, under a controlled comparison, does guided search actually dominate linear CoT? Second, if not, do the two methods succeed on complementary instances? Third, can this complementarity be explained by pre-generation structural properties of the input? Finally, can those structural properties be used to route between CoT and search before any reasoning trace is generated?

To answer these questions, we instantiate search-based reasoning using A* search over natural-language reasoning traces. A* maintains a frontier of partial reasoning states and expands them according to a priority score $f(n) = g(n) + h(n)$, where $g(n)$ captures the cost already incurred and $h(n)$ estimates the remaining cost to a correct answer. We use A* not because it is the only possible search strategy, but because it makes the ingredients of search explicit: frontier ordering, accumulated cost, heuristic guidance, branching budget, and stopping rule. This lets us use A* as a controlled abstraction for studying the value of guided exploration in LLM reasoning.

Figure 1 illustrates the distinction. For the question “*Is shrimp scampi definitely free of plastic?*”, CoT follows a chronological preparation chain: harvesting, cleaning, cooking, ingredients, and regulation. Each step is locally plausible, but the trace misses the decisive biological premise that shrimp may already contain microplastics. A* instead prioritizes this evidence branch and then checks whether preparation removes the contamination. The point is not that A* always behaves better; rather, this example shows the kind of topology where search can help: the useful evidence branch is not the first path a linear explanation follows.

Our findings are consistent across several analyses. Aggregate accuracy is inconclusive: A* wins on StrategyQA and HotpotQA but not on LogicQA. At the instance level, however, CoT and A* are strongly complementary. We explain this complementarity through a Reasoning Topology Framework: as branching complexity increases, A* becomes more useful; when the problem is short and linear, CoT is sufficient. Cross-dataset topic modelling independently recovers the same pattern. Finally, a pre-generation topology router outperforms routers that inspect generated traces, showing that the key signal is problem structure rather than trace fluency.

Why A*? Our goal is not to argue that A* is the universally best search algorithm for LLM reasoning. Rather, we use A* as a controlled abstraction for *heuristic-guided frontier search*. This choice is useful because it exposes the minimal ingredients that distinguish guided search from linear CoT: a frontier of partial reasoning states, an accumulated path cost, an estimate of remaining reasoning cost, a branching budget, and a stopping rule. Simpler traversals such as DFS and BFS isolate only one aspect of search: DFS tests depth-first commitment, while BFS tests unguided breadth. Beam search and Tree-of-Thoughts add evaluator-based pruning, but often entangle local scoring, decomposition prompts, and breadth choices. Graph-of-Thoughts and MCTS-style methods introduce still richer design choices such as graph composition, rollout policy, value estimation, and exploration constants. These methods are important baselines, but they are less suitable as the primary diagnostic object because improvements or failures become harder to attribute.

A* provides a compact middle ground. It selects the next partial trace by minimizing

$$f(n) = g(n) + h(n),$$

Motivating Example: “Is shrimp scampi definitely free of plastic?” (Gold: FALSE)	
✗ Chain-of-Thought (Linear Generation) <i>Trace explores chronological preparation steps, dismissing the biological premise due to early linear commitment.</i> Step 1: Shrimp harvesting: Shrimp are typically caught in the wild or farmed... While it’s possible for microplastics to be present in seawater, there is no conclusive evidence that shrimp themselves contain significant amounts of plastic. Step 2: Processing and preparation: After harvesting, shrimp are usually cleaned... this process doesn’t necessarily involve the use of plastics. Step 3: Cooking and serving: ...The cooking process itself does not introduce plastic into the dish. Step 4: Ingredient sourcing: ...there is no direct evidence that these contaminants are significant enough to be considered a problem. Step 5: Regulatory oversight: ...the overall likelihood of plastic contamination in shrimp scampi is low. Ans: Therefore: Final Answer: YES (Incorrect)	✓ A* Search (Heuristic-Guided) <i>Heuristic prioritizes high-relevance nodes, bypassing chronological steps to aggregate non-linear evidence.</i> Step 1: Many types of seafood, including shrimp, have been found to have microplastics in their digestive systems and tissues. Step 2: Shrimp scampi preparation typically involves cooking shrimp in butter and garlic, but it does not involve processes that would remove or clean microplastics from the shrimp. Ans: Pred: FALSE (Correct)

Figure 1: A qualitative comparison of reasoning trajectories. Standard CoT commits to a linear analysis of food preparation, arriving at a locally coherent but globally incorrect conclusion. In contrast, A* search utilizes its domain-aware heuristic to prioritize the high-relevance biological premise (microplastic bioaccumulation), bypassing exhaustive chronological steps to arrive at the correct conclusion.

where $g(n)$ measures the cost already incurred and $h(n)$ estimates the remaining cost to a correct answer. Thus A* directly operationalizes the question we wish to study: when does a problem benefit from spending additional computation on guided exploration rather than committing to a single linear trace? In this sense, A* is not treated as an optimal planner with classical admissibility guarantees over language traces; it is used as an interpretable priority-search policy for studying the value and failure modes of guided exploration.

Contributions.

1. We provide a controlled comparison of linear CoT and guided A* search across three reasoning benchmarks, showing that search does not uniformly dominate and that aggregate benchmark accuracy hides strong instance-level complementarity.
2. We introduce a Reasoning Topology Framework that characterizes problems using pre-generation structural features and shows that branching complexity predicts when A* outperforms CoT.
3. We validate the topology explanation through cross-dataset topic modelling and topology-aware routing, showing that pre-generation problem structure is more predictive than post-generation trace inspection.
4. We analyze why search helps and fails through fluency–correctness asymmetry, overthinking failures, and small-model capability amplification, yielding a practical account of when search should be used.

Together, these results support a single thesis: search is not a universally stronger replacement for CoT. It is a topology-sensitive computation strategy. The practical question is therefore not simply whether an LLM should reason with CoT or with search, but which reasoning topology a given problem instantiates, and which inference strategy matches that topology.

2 Related Work

Chain-of-thought and structured reasoning. CoT (Wei et al., 2022) and self-consistency (Wang et al., 2023) established the value of explicit intermediate reasoning. ToT (Yao et al., 2024) and GoT (Besta et al., 2024) extend to richer structures. CoT-decoding (Wang and Zhou, 2024) shows that reasoning paths exist in top- k alternative tokens without prompting. A recent empirical study finds CoT is not universally beneficial (Meincke et al., 2025); our result is the search-side complement—identifying precisely the structural conditions under which search helps. Our work differs in three respects: we use A* as an explicit priority-search abstraction, we empirically study when search hurts rather than only when it helps, and we connect search effectiveness to pre-generation topology features.

Search-augmented reasoning. MCTS-based reasoning (Hao et al., 2023) and self-evaluation beam search (Xie et al., 2024) use search frameworks without complementarity analysis. LATS (Zhou et al., 2024) unifies reasoning, acting, and planning via MCTS. MCTSr (Zhang et al., 2024) applies MCTS with self-refinement to mathematical Olympiad problems. Coconut (Hao et al., 2024) enables implicit BFS through continuous latent states—our work is the discrete, interpretable counterpart. rStar-Math (Guan et al., 2025) demonstrates search helps SLMs on mathematics; we confirm this in QA and identify the capability–topology interaction that determines the magnitude of the gain (§8.2).

Over-exploration and search failures. The Fetch framework (Zhang et al., 2025a) identifies redundant-state over-exploration; this corresponds to our step repetition failure mode. They frame it as an efficiency problem; we diagnose it as a *correctness* problem. The overthinking survey (Zhang et al., 2025b) documents the phenomenon in long-CoT models; CoT-Valve (Ma et al., 2025) addresses it via length-compressible tuning. Our analysis provides a per-instance quantification of overthinking failures in search-based reasoning, complementing prior work that primarily studies over-exploration as an efficiency problem.

Routing and adaptive computation. Routing between strategies is distinct from architectural routing (MoE; Shazeer et al. 2017). A recent study (Meincke et al., 2025) shows CoT is not universally beneficial; our result is the search-side mirror. Most adaptive inference methods use signals produced during or after generation: uncertainty, entropy, verifier scores, or trace-level judgments. Our setting asks whether the strategy can be chosen before generation, using only the structural properties of the input. This distinction matters because trace-level judgments can be confounded by fluency.

LLM-based evaluation. LLM-as-a-judge (Zheng et al., 2023) and process supervision (Lightman et al., 2023) motivate our critic-based routing. The Fluency–Correctness Asymmetry explains why critic-based routing reaches a ceiling. Our routing problem is also distinct from architectural mixture-of-experts routing: the router selects between fundamentally different inference-time reasoning algorithms, not between model components.

3 Reasoning Strategies and Experimental Setup

This section defines the inference strategies and experimental protocol used in our diagnostic study. We separate four components: (i) the space of reasoning traces, (ii) CoT as a linear sampling policy over that space, (iii) A* as a guided frontier-search policy, and (iv) the pre-generation structural features used to characterize problem topology.

3.1 Notation and Problem Setup

Let $Q = (q, C, O)$ denote a reasoning problem, where q is the question, C is an optional context passage, and $O = \{o_1, \dots, o_K\}$ is the set of candidate answers for multiple-choice tasks or the space of free-form answers. A reasoning trace $\tau = (s_1, s_2, \dots, s_T)$ is an ordered sequence of natural-language reasoning steps, and a prediction function $\phi(\tau) \in O$ extracts the final answer from the trace. Let M_θ denote the language model with parameters θ .

This notation lets us compare CoT and A* as two different ways of exploring the same latent space $\mathcal{T}$ of possible reasoning traces. CoT samples a single trace from this space. A* explicitly maintains a frontier of partial traces and decides which partial trace to expand next.

3.2 Chain-of-Thought as Linear Reasoning

In CoT prompting, the model generates a reasoning trace autoregressively:

$$\tau^{\text{CoT}} = (s_1, s_2, \dots, s_T), \quad s_t \sim M_\theta(\cdot \mid q, C, O, s_{1:t-1}), \quad (1)$$

where each step s_t is conditioned on all prior steps $s_{1:t-1}$. Generation terminates when the model emits a terminal answer pattern (for example, “The answer is [X]”). The final prediction is $\hat{y}^{\text{CoT}} = \phi(\tau^{\text{CoT}})$.

Limitations of linear generation. The CoT generation in Eq. (1) defines a single path through the space of possible reasoning traces. CoT explores exactly one element $\tau^{\text{CoT}} \in \mathcal{T}$, with no mechanism to: (i) evaluate the quality of intermediate steps before committing to them, (ii) explore alternative branches when a step appears unproductive, or (iii) aggregate evidence from multiple reasoning paths. This is precisely why CoT is vulnerable to premature commitment: once a misleading intermediate step is generated, all subsequent steps inherit its assumptions. This limitation is not equally harmful for all inputs. It matters precisely when the problem requires maintaining alternatives or revising an early premise. This observation motivates our later topology features.

Naive alternatives such as breadth-first search (BFS) or depth-first search (DFS) over $\mathcal{T}$ address only part of this limitation. BFS explores alternatives but can incur large memory costs; DFS can pursue arbitrarily deep unproductive branches; beam search depends heavily on local scoring. We use A* as a controlled priority-search abstraction for studying guided exploration while keeping the frontier, cost, heuristic, and budget explicit.

3.3 A* Search over Reasoning Traces

We formalise reasoning as graph search over partial natural-language traces. The search space is a tree $\mathcal{G} = (\mathcal{S}, \mathcal{E})$, where each node $n \in \mathcal{S}$ represents a partial reasoning state:

$$n = (\tau_n, d_n), \quad \tau_n = (s_1, \dots, s_{d_n}), \quad (2)$$

where τ_n is the partial trace and $d_n = |\tau_n|$ is its depth. The root node has $\tau_{n_0} = \emptyset$ and $d_{n_0} = 0$. An edge $(n, n') \in \mathcal{E}$ exists when $\tau_{n'} = \tau_n \oplus s$ for some generated reasoning step s .

Successor generation. Given a node n , we generate $N_c = 3$ candidate successor steps by prompting the base language model to extend the partial reasoning trace from different inference angles, such as supporting evidence, counterargument, option-wise elimination, or factual lookup:

$$\{s^{(j)}\}_{j=1}^{N_c} = \text{GenerateSteps}(M_\theta, q, C, O, \tau_n, N_c). \quad (3)$$

Each candidate yields a successor node n'_j with $\tau_{n'_j} = \tau_n \oplus s^{(j)}$ and $d_{n'_j} = d_n + 1$.

Goal condition. A node is a goal node if its most recent step contains a terminal answer pattern:

$$\text{IsGoal}(n) = \mathbf{1} [\exists p \in \mathcal{P} : p \subseteq s_{d_n}], \quad (4)$$

where $\mathcal{P}$ includes patterns such as “The answer is [A/B/C/D]” for multiple-choice tasks or an explicit final-answer marker for free-form QA.

Priority score. A* maintains an open list of frontier nodes sorted by

$$f(n) = g(n) + h(n), \quad (5)$$

where $g(n)$ is the accumulated path cost and $h(n)$ is the estimated remaining cost to a correct answer. In the implementation used for the main experiments, $g(n)$ tracks accumulated reasoning cost and depth, while $h(n)$ is produced by a critic model that evaluates the partial state.

Algorithm 1 Budgeted A* search over reasoning traces**Require:** question q , context C , options O , model M_θ , critic M_{critic} , expansion budget B , branching factor N_c

```

1: Initialise root  $n_0 = (\emptyset, 0)$ 
2: Initialise priority queue OPEN  $\leftarrow \{n_0\}$  with priority  $f(n_0) = g(n_0) + h(n_0)$ 
3: Initialise goal set  $\mathcal{G}_{\text{goal}} \leftarrow \emptyset$ 
4: for  $b = 1$  to  $B$  do
5:   Pop  $n = \arg \min_{m \in \text{OPEN}} f(m)$ 
6:   if IsGoal( $n$ ) then
7:     Add  $n$  to  $\mathcal{G}_{\text{goal}}$ 
8:     continue
9:   end if
10:  Generate successors  $\{n'_j\}_{j=1}^{N_c}$  using  $M_\theta$ 
11:  for each successor  $n'_j$  do
12:    Score  $n'_j$  using  $M_{\text{critic}}$ 
13:    Set  $h(n'_j) = 1 - \text{score}(n'_j)$  and compute  $g(n'_j)$ 
14:    Push  $n'_j$  into OPEN with priority  $f(n'_j) = g(n'_j) + h(n'_j)$ 
15:  end for
16: end for
17: return highest-scoring goal node if available; otherwise return the best frontier node

```

Heuristic evaluation. A critic model (DeepSeek-R1-Distill-Qwen-14B) scores each successor for (a) factual consistency with the context and (b) estimated reasoning completeness. The critic returns a score in $[0, 1]$, which we convert to

$$h(n) = 1 - \text{score}(n). \quad (6)$$

Higher-quality states therefore receive lower estimated remaining cost. This keeps the search procedure close to classical A*: states estimated to be nearer to a correct answer are expanded earlier.

Termination and answer aggregation. Search terminates when a node reaches a definitive answer or when a budget of $B = 15$ expansions is exhausted. In the main experiments, the highest-scoring terminal node is returned. We retain the following weighted goal-node voting rule as an implementation variant and auxiliary analysis:

$$\hat{y}^{A*} = \arg \max_{y \in O} \sum_{n \in \mathcal{G}_{\text{goal}}: \phi(\tau_n)=y} \frac{1}{g(n) + \epsilon}, \quad (7)$$

where $\epsilon > 0$ is a small constant for numerical stability. This variant formalizes the intuition that shorter and lower-cost reasoning paths should count more strongly, while the main experiments use the critic score as the primary selection signal.

3.4 Heuristic Scoring and Approximate Admissibility

A heuristic $h(n)$ is admissible if $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true remaining cost. In classical planning, admissibility ensures A* finds an optimal solution. In LLM reasoning, however, states are natural-language traces and the heuristic is model-derived; perfect admissibility cannot be assumed. We therefore analyse sufficient approximate conditions under which the critic-generated heuristic behaves like a conservative or near-admissible guide.

Proposition 1 (Sufficient conditions for conservative heuristic guidance). *Let $h(n) = 1 - \text{score}(n)$ be the critic-generated heuristic. Then $h(n)$ can act as a conservative guide under any of the following conditions:*

Case 1 Monotone calibrated scoring. *If critic scores are monotone in true reasoning progress and calibrated so that $1 - \text{score}(n)$ lower-bounds the remaining normalized cost, then $h(n)$ is conservative in the normalized search space.*

Case 2 Conservative estimation. *If $\mathbb{E}[\text{score}(n)] \leq \text{score}^*(n)$, the critic systematically underestimates true progress. The resulting heuristic is conservative in expectation.*

Case 3 Bounded overestimation. If $|\text{score}(n) - \text{score}^*(n)| \leq \epsilon$, then search is approximately admissible in the sense that occasional critic error is bounded.

Empirical validation. We annotate 500 reasoning states (100 per dataset for StrategyQA and HotpotQA, 300 for LogicQA). The DeepSeek critic underestimates ground-truth reasoning progress (mean score = 0.43 vs. human-assessed = 0.58; paired t -test $p < 0.001$), satisfying the conservative-estimation case. The overestimation rate is 12.3% of nodes with maximum $\epsilon_{\max} = 0.14$, providing an empirical bound for the bounded-overestimation case. These conditions are sufficient but not necessary: A* can still be useful when the heuristic is occasionally wrong, especially because the expansion budget regularises search.

Auxiliary conservative heuristic. Appendix B describes an alternative depth-coverage heuristic that illustrates how conservative search guidance can be constructed when minimum-hop assumptions and entity-coverage terms are available. The main experiments use the critic-based heuristic above.

3.5 Benchmarks and Data

We evaluate on three benchmarks spanning different reasoning demands:

- **LogicQA** (Liu et al., 2020): 651 formal logic questions (4-way multiple choice).
- **StrategyQA** (Geva et al., 2021): 500 commonsense questions requiring implicit multi-step reasoning (yes/no).
- **HotpotQA** (Yang et al., 2018): 1,000 multi-hop factual questions (free-form answer).

Fair comparison protocol. Both A* and CoT are run on exactly the same questions with exactly the same model. An earlier experimental setup compared A* with LLaMA 3.1-8B against CoT with Qwen-0.5B, confounding strategy differences with a $16\times$ parameter gap. We therefore filled all CoT gaps and produced a fully balanced $2,151 \times 2$ comparison.

Generation parameters. All experiments use Ollama v0.5 with temperature $\tau = 0.7$, top- $p = 0.9$, and maximum generation length of 512 tokens per reasoning step. For self-consistency, we sample 5 traces with $\tau = 0.9$ to encourage diversity. For Tree-of-Thoughts, we use BFS with 3 candidates per step and 5-step depth. A* uses $N_c = 3$ candidate expansions per node and a budget of $B = 15$ expansions. The critic model uses $\tau = 0.3$ for more deterministic scoring. Exact prompts are provided in Appendix F.

3.6 Comparison Methods

We compare five reasoning strategies, all using LLaMA 3.1-8B via Ollama:

- **CoT**: Zero-shot chain-of-thought with “Let’s think step by step” (Wei et al., 2022).
- **Self-Consistency (SC)**: Majority vote over 5 sampled CoT traces (Wang et al., 2023).
- **Tree-of-Thoughts (ToT)**: BFS over 3 candidates per step and 5-step depth (Yao et al., 2024).
- **A* Search**: The budgeted priority-search procedure described above.
- **Oracle**: Per-instance best of A* and CoT, used only as an upper bound.

3.7 Reasoning Topology Features

We define 13 structural features for each question, computed without running any model. These features enable routing decisions before inference begins and form the basis of the topology analysis. The full feature set is described in Table 1.

Table 1: Reasoning topology features, all computed from raw question and context text without LLM inference.

Feature	Symbol	Description
Question length	q_{len}	Token count of question
Context length	c_{len}	Token count of full context
Context sentences	c_{sent}	Number of sentences
Logical connectives	conn	Count of <i>if, then, therefore, because, unless</i>
Negation markers	neg	Count of <i>not, no, never, neither</i>
Comparatives	cmp	Count of <i>more, less, better, worse, than</i>
Question marks	$q?$	Number of question marks
Hop count	hop	Estimated reasoning hops (dependency parse)
Contextual density	con	Entities per sentence in context
Distractors	dest	Number of candidate options (MCQ)
Branching coeff.	bc	$c_{\text{sent}} \times \text{hop} / q_{\text{len}}$
Reasoning depth	ℓ	Maximum dependency parse depth
Number of options	n_{opt}	Answer options available

Table 2: Aggregate accuracy (%) across three benchmarks with 95% Wilson confidence intervals. Best per dataset in **bold**. Oracle = per-instance best of A* and CoT. [†]McNemar’s test (A* vs. CoT): $p < 0.001$ on StrategyQA, $p = 0.61$ on LogicQA, $p = 0.048$ on HotpotQA.

Method	StrategyQA ($n=500$)	LogicQA ($n=651$)	HotpotQA ($n=1000$)
CoT	64.52 $\pm$ 4.2	45.78 $\pm$ 3.8	76.80 $\pm$ 2.6
Self-Consistency	71.22 $\pm$ 4.0	46.10 $\pm$ 3.8	78.21 $\pm$ 2.6
Tree-of-Thoughts	72.24 $\pm$ 3.9	46.21 $\pm$ 3.8	78.45 $\pm$ 2.5
A* Search [†]	74.80 $\pm$ 3.8	44.85 $\pm$ 3.8	80.10 $\pm$ 2.5
Oracle (A* $\cup$ CoT)	84.59 $\pm$ 3.2	59.29 $\pm$ 3.8	94.20 $\pm$ 1.4

3.8 Complementarity Diagnostic

The empirical study is organized around a simple diagnostic question: after CoT and A* are run on the same input with the same base model, do they fail on the same examples or on different examples? We use *complementarity* to mean that each strategy solves a non-trivial subset of instances that the other strategy misses. Formally, every instance falls into one of four cells: *both correct*, *CoT only*, *A* only*, or *both wrong*. The *oracle* score, which selects the correct method whenever either CoT or A* is correct, measures the maximum possible benefit of perfect routing.

This diagnostic determines the rest of the paper. If one method dominates in aggregate and the oracle adds little, then there is no meaningful routing problem. If the *CoT-only* and *A*-only* cells are both substantial, then the methods implement different useful computations, and the central question becomes explanatory: what properties of the input make one computation preferable to the other? Sections 4.1 and 4.2 establish this complementarity empirically. Sections 5.1–5.2 then test whether it is explained by reasoning topology. Section 6.1 finally asks whether topology can be used for pre-generation strategy selection.

4 From Aggregate Inconsistency to Complementarity

We first establish that the comparison between CoT and A* is not resolved by aggregate accuracy alone. This section has two steps. Section 4.1 tests whether one strategy uniformly dominates. Section 4.2 then examines the instance-level outcome cells defined in Section 3.8. The purpose is to determine whether there is a real strategy-selection problem to explain.

4.1 Aggregate Performance: The Inconsistency

Table 2 presents aggregate results. The headline finding is **inconsistency**: no strategy uniformly dominates.

Table 3: Instance-level agreement (% of questions).

	StrategyQA	LogicQA	HotpotQA
Both correct	54.7	31.3	62.7
CoT only	9.8	14.4	14.1
A* only	20.1	13.5	17.4
Both wrong	15.4	40.7	5.8

A* leads on StrategyQA (+10.3 pp, $p < 0.001$) and HotpotQA (+3.3 pp, $p = 0.048$) but the LogicQA gap (-0.9 pp, $p = 0.61$) is not statistically significant. SC and ToT consistently fall between CoT and A*, suggesting they capture part but not all of the value of guided search. The oracle row reveals substantial **complementarity**: on HotpotQA, 94.2% of questions are answered correctly by at least one method versus 80.1% for A* alone—a 14.1 pp gap that represents the theoretical ceiling for a perfect router.

This result rules out the simplest hypothesis: that search is uniformly better because it explores more reasoning paths. If that were true, A* should dominate across all three benchmarks. Instead, the sign and size of the gain vary by benchmark, suggesting that the relevant variable is not search budget alone but the structure of the inference required by each instance.

4.2 Instance-Level Complementarity

Table 3 presents per-instance outcome matrices.

On all three datasets, 28–32% of questions are answered correctly by *exactly one* method, confirming genuinely complementary failure patterns. LogicQA’s high “both wrong” rate (40.7%) reflects the intrinsic difficulty of formal logic for current LLMs.

The oracle gains show that the aggregate inconsistency is not merely noise. CoT and A* solve partially disjoint subsets of problems. Therefore, the key scientific question becomes explanatory: what distinguishes A*-only instances from CoT-only instances? The next section tests the central hypothesis of the paper: that this distinction is largely governed by the topology of the reasoning problem.

5 Explaining Complementarity Through Topology

Having established complementarity, we now ask what explains it. Our hypothesis is that search helps when the input has a branching reasoning topology: several premises, entity paths, hypotheses, or temporal steps must be preserved before the answer can be selected. Conversely, CoT should be sufficient when the problem admits a compact linear inference. We test this hypothesis in two ways: first through structural topology features, and then through an independent cross-dataset topic analysis.

5.1 Reasoning Topology: Four Cluster Types

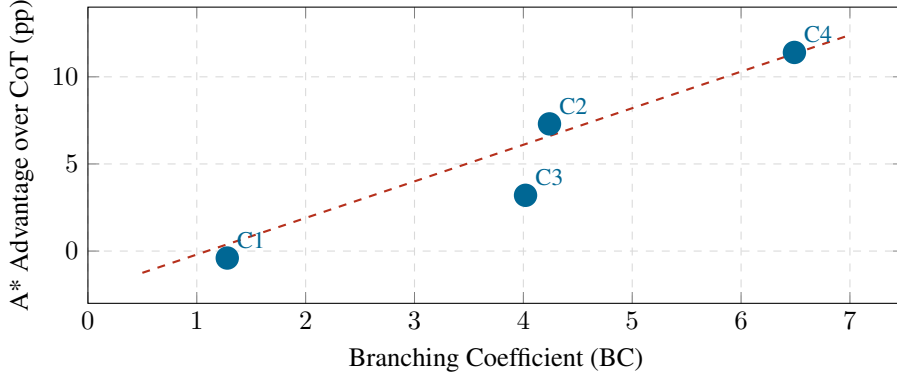
We use *topology* to refer to the observable structure of the reasoning problem: how much context must be inspected, how many candidate premises are available, how many logical or temporal dependencies must be maintained, and whether a single linear path is likely to be sufficient. These features describe the input before any model output is generated.

We fit K-means ($k=4$) on the 13 topology features across all 2,151 questions. Table 4 shows the resulting clusters.

As branching complexity increases ($C1 \rightarrow C4$), A*’s advantage grows from -0.4 pp to $+11.4$ pp. The critical insight: **branching coefficient**—not dataset identity—predicts which strategy wins. Figure 2 visualises this monotonic relationship. The clusters should not be read as dataset labels: each cluster contains instances from multiple benchmarks. The result therefore suggests that the CoT/search boundary cuts across benchmark boundaries and is better explained by problem structure than by dataset name alone.

Table 4: Reasoning topology clusters. BC = branching coefficient. Both% = fraction where both strategies are correct.

Cluster	Profile	BC	A*	CoT	Both%	Interpretation
C1	Linear, short	1.28	81.7	82.1	76.4	CoT sufficient
C2	Mid-branch	4.24	73.5	66.2	57.8	Selection critical
C3	Complex, noisy	4.02	52.3	49.1	38.6	Both often fail
C4	Deep-branch	6.49	69.8	58.4	47.2	A* advantage largest

Figure 2: A* advantage grows monotonically with branching coefficient (dashed: linear fit, $R^2=0.94$). Each point is one topology cluster from Table 4.

Prescriptive decision rule.

- **C1** (linear): Use CoT—search overhead is unnecessary.
- **C2** (mid-branch): Route adaptively—largest marginal gains.
- **C3** (complex): Neither reliable—invest in retrieval augmentation.
- **C4** (deep-branch): Use A*—search is essential.

5.2 Topic Modelling Validates the Topology Explanation

The topology clusters use hand-crafted structural features. To check that the conclusion is not an artefact of those features, we perform an independent cross-dataset topic analysis over the raw question texts. The role of this analysis is validation: if the topology story is correct, content categories that require multi-hop or premise-scanning structure should favor A*, while categories dominated by local inference should favor CoT.

We apply NMF on TF-IDF representations of all 2,151 question texts ($\text{max_df}=0.80$, $\text{min_df}=3$, $\text{max_features}=5,000$, $\text{ngram_range}=(1, 2)$, $k=8$, $\text{random_state}=42$). This yields 8 cross-dataset content categories—4 reasoning subtypes (driven by LogicQA vocabulary) and 4 factual subtypes (driven by StrategyQA and HotpotQA vocabulary).

Topic quality validation. We select $k=8$ via NPMI coherence (peaks at $k=8$: 0.127; vs. $k=6$: 0.103; $k=10$: 0.118). The A*-dominance finding is robust: 79.2% at $k=6$, 82.1% at $k=10$. Bootstrap stability (10-fold): 87.3% mean topic-assignment agreement (std. 2.1%).

Key finding. The important finding is not simply that A* wins in five topics. It is that the A*-dominant topics share the same structural signature identified by the topology analysis: multi-hop chaining, premise scanning, temporal ordering, and entity resolution. Conversely, the CoT-dominant topics are those where the answer typically follows from one local comparison or option selection.

Table 5: Per-topic A* vs. CoT accuracy (%) across all three datasets. Best per cell in **bold**. W = winner.

Topic	Cov.	LogicQA (A*/CoT)	StrategyQA (A*/CoT)	HotpotQA (A*/CoT)	W
content_domain_factual	43.7	38/38	75/56	81/74	A*
assumption_analysis	13.9	47/45	82/50	—	A*
temporal_event	9.8	40/40	79/75	80/66	A*
evaluate_arg_eff.	9.1	43/41	78/76	62/62	A*
comparison_factual	8.7	44/ 50	71/ 73	71/59	CoT
statement_strength	5.3	44/ 45	33/ 67	—	CoT
biography_factual	4.9	—	62/58	87/82	A*
best_option_sel.	4.6	45/ 55	100/75	100/100	CoT

Table 6: Summary of topic-level dominance.

	Topics	Coverage
A* wins	content_domain_factual, assumption_analysis, temporal_event, evaluate_arg_effectiveness, biography_factual	81.4%
CoT wins	comparison_factual, statement_strength_eval, best_option_selection	18.6%

A* wins in 5 of 8 categories covering 81.4% of the corpus. Figure 3 visualises the coverage distribution. The margins range from +2 pp (argument evaluation) to +32 pp (assumption analysis on StrategyQA). All A*-dominant categories share a structural property: they require chaining two or more interdependent facts—entity lookups, temporal chains, premise scanning, and biographical identity resolution. The CoT-dominant categories are all single-step inference tasks.

Why A* dominates these categories.

- *content_domain_factual* (43.7%): 2–3 knowledge hops. A*’s graph expansion follows each hop methodically; CoT shortcuts and hallucinates entities. Best gap: +19 pp.
- *assumption_analysis* (13.9%): Requires scanning *all* premises to find the implicit gap. A*’s exhaustive node expansion is a natural full-premise scan. Best gap: +32 pp.
- *temporal_event* (9.8%): Multi-hop temporal chains. A*’s sequential hop expansion maps directly. Best gap: +14 pp.
- *biography_factual* (4.9%): Person-identity chains requiring entity disambiguation. A*’s branching explores alternative resolutions. Best gap: +5 pp.

Why CoT wins on 18.6%. The three CoT-dominant categories—*comparison_factual*, *statement_strength_eval*, *best_option_selection*—share a common structure: the answer emerges from a *single reasoning step*. When only one hop separates question from answer, A*’s branching adds overhead without benefit.

6 Topology-Aware Routing

The instance-level complementarity in Section 4.2 creates a routing problem: for each new input, choose whether to run CoT or A* before knowing which one will be correct. The topology analysis in Section 5 suggests a stronger version of this problem: routing should not be treated only as post-hoc trace adjudication, because the input structure itself may already indicate which computation is appropriate.

We compare two families of routers. *Output-based routers* inspect samples or traces produced by the model. Semantic entropy computes uncertainty over self-consistency samples and routes high-entropy cases to A*. LLM-as-critic routing invokes DeepSeek-R1-Distill-Qwen-14B to adjudicate when CoT and A* disagree. *Pre-generation routers*, by contrast, use only structural features of the input. The feature-based random forest uses surface features such as length, negation,

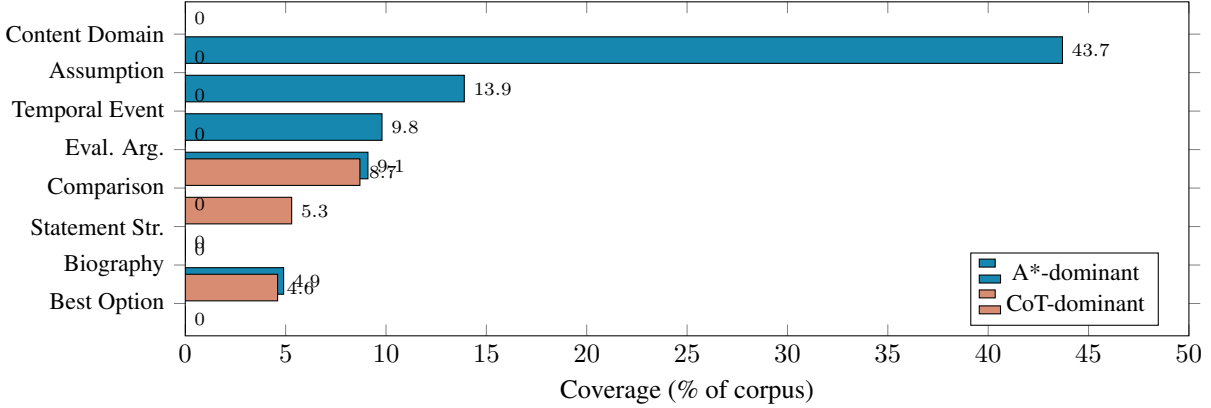


Figure 3: Topic coverage: A*-dominant (blue, 81.4%) vs. CoT-dominant (red, 18.6%). NMF $k=8$ across all 2,151 questions.

Table 7: Router comparison. Gap recovery = percentage of oracle gap closed.

Router	Input	StrategyQA	LogicQA	Gap Rec.
A* only	—	74.80	44.85	—
CoT only	—	64.52	45.78	—
Semantic Entropy	Traces	74.37	47.47	14.8%
LLM-as-Critic	Traces	74.19	47.16	12.4%
Random Forest	Features+labels	76.25	48.20	18.2%
Topology Router	13 features	77.35	49.01	26.1%
Oracle	—	84.59	59.29	100%

and connectives. The topology router uses the full 13-feature topology representation from Table 1. Thus the routing experiment directly tests whether the complementarity observed earlier is predictable before reasoning traces are generated.

6.1 Routing: Topology Beats Trace Inspection

Router implementations. The **Topology Router** is a gradient-boosted decision tree (XGBoost; 100 estimators, max depth 6, learning rate 0.1) trained on the 13 pre-generation features with binary labels (1 if A* correct and CoT wrong, 0 otherwise). The **Random Forest** baseline uses a simpler hand-crafted feature set. **Semantic Entropy** and **LLM-as-Critic** are the output-based baselines described above. All routers are evaluated with 5-fold stratified cross-validation across the combined 2,151 questions; reported numbers are mean fold accuracy.

The topology router—which sees *zero model outputs*—recovers **26.1%** of the oracle gap, outperforming all output-based routers. This inverts the presumed solution direction: the bottleneck is recognising problem structure *before* inference begins, not evaluating trace quality. This result is counterintuitive. One might expect a router that observes the completed reasoning traces to have more information than a router that only sees the input. Instead, trace-based routers underperform because their signals are contaminated by fluency and narrative coherence. The topology router succeeds because it predicts which computation should be run before either computation has produced a persuasive trace.

Feature importance. The most predictive features are c_{len} (0.208), q_{len} (0.194), c_{sent} (0.113), and con (0.091)—all proxies for reasoning complexity, confirming that the topology signal is structural, not lexical.

Table 8: Fluency–Correctness Asymmetry on disagreement cases.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where more fluent	62.0%	38.0%
More-fluent but wrong		38.0%

6.2 Fluency–Correctness Asymmetry

We next ask why output-based routers fail to close more of the oracle gap. We measure fluency via perplexity-based readability on all cases where A* and CoT disagree (Table 8).

Although A* has the higher average fluency score overall, a substantial minority of disagreement cases exhibit the dangerous pattern for routing: CoT is the more fluent trace but is wrong. These cases create a ceiling for routers that equate trace readability with correctness. A fluency-based router would systematically misroute such cases. Human evaluation corroborates: on StrategyQA cases where A* is correct, A* achieves coverage 2.71 vs. CoT 1.63 (3-point scale), yet CoT relevance remains at 2.47—incorrect traces stay on-topic and look reasonable.

7 Qualitative Mechanisms Behind the Topology Effect

The topology and topic analyses identify where A* and CoT differ. We now examine representative qualitative examples to clarify why these differences arise. The examples are not used as evidence by themselves; rather, they illustrate the mechanisms suggested by the aggregate analyses: premature commitment in CoT, hypothesis preservation in A*, and overthinking when search is applied to linear problems.

7.1 Patterns of A* Success and CoT Failure

Non-linear evidence aggregation. These are problems where the correct answer requires synthesising multiple independent facts that CoT processes sequentially, losing sight of the global picture. The shrimp scampi example in Figure 1 is a canonical instance. CoT proceeds through harvesting, processing, cooking, ingredient sourcing, and regulation, and each step locally concludes that no substantial plastic contamination is introduced. The trace is coherent but misses the biological premise: shrimp can contain microplastics before cooking. A* instead identifies the microplastic bioaccumulation premise early and then checks whether scampi preparation removes it. This matches the topology story: the useful reasoning path is not the most obvious chronological path.

StrategyQA: Non-linear Evidence Aggregation

Question: Is shrimp scampi definitely free of plastic? **Gold:** No.

CoT failure: follows the preparation pipeline and repeatedly concludes that no preparation stage introduces plastic.

A* success: jumps to the biological evidence branch—microplastics in seafood—and then verifies that cooking does not remove them.

Hypothesis elimination. Some LogicQA questions require systematic elimination of answer options. CoT can identify the broad logical form but then map it to the wrong option because of surface similarity. A* is better matched to this setting because branches can separately evaluate candidate options.

LogicQA: Hypothesis Elimination

Question pattern: Which option is most similar to the argument: “All landscape rooms can see the landscape, but Li Wenbing’s family cannot see the landscape. Therefore, Li Wenbing’s family is not a landscape room.”

CoT failure: recognises a negative inference pattern but chooses an option with surface-level similarity.

A* success: explores candidate options independently and identifies the one preserving the modus-tollens structure.

Multi-hop entity chaining. HotpotQA questions often require preserving multiple evidence chains until the right entity or role is identified. CoT may correctly identify an intermediate entity but then retrieve a salient wrong fact from parametric memory. A* helps by keeping alternative continuations alive.

HotpotQA: Multi-hop Chaining

Question: What government position was held by the woman who portrayed Corliss Archer in the film *Kiss and Tell*? **Gold:** Chief of Protocol.

CoT failure: identifies Shirley Temple but confuses her government roles and answers with a more salient ambassadorial position.

A* success mode: explores multiple evidence paths for Shirley Temple’s career before committing.

7.2 Patterns of CoT Success and A* Failure

Direct factual reasoning. When the answer follows straightforwardly from well-known facts, A*’s search overhead can introduce unnecessary branching and lead the model away from the direct solution. This corresponds to C1 in the topology analysis: linear, short questions where CoT is sufficient. These are not cases where search fails because it is weak; they are cases where the problem does not provide useful branches to explore. The additional search space is mostly distractor space.

StrategyQA: Direct Factual Reasoning

Question: Is a Boeing 737 cost covered by *Wonder Woman* (2017) box office receipts? **Gold:** Yes.

CoT success: retrieves the rough magnitude of a Boeing 737 cost and the film’s box office receipts, then compares them directly.

A* failure: over-interprets the wording as a financial-transaction or cost-accounting question and explores irrelevant branches.

Fluent world knowledge and argumentation. Some questions are best solved by maintaining a single coherent argumentative frame. A*’s fragmented partial traces can lose global narrative consistency, especially when the correct answer depends on a smooth integration of commonsense or argument structure rather than exhaustive branching.

CoT Success through Coherent Argumentation

Pattern: Questions where the model must preserve the global argumentative frame across the answer options.

CoT success: maintains a coherent single narrative and compares options within that frame.

A* failure: anchors on one branch or option too early and does not reconstruct the global argument.

7.3 Fine-Grained A* Failure Modes

The aggregate taxonomy in Section 8.1 shows that most A* failures are overthinking. Manual inspection reveals three recurring forms: semantic over-refinement, where the model turns a direct fact into a technical distinction; associative detours, where semantically related facts are chained into a spurious argument; and budget dilution, where repeated or near-duplicate steps prevent the search from reaching the decisive comparison.

Failure mode 1: Semantic over-refinement. For straightforward factual questions, A*’s multi-step decomposition can introduce unnecessary complexity. A representative example is the question of whether Disney has an ice princess. The direct answer is available through Elsa, but A* can continue into irrelevant distinctions about queen versus princess status, franchise naming, and semantic interpretation.

Table 9: Representative qualitative failure modes illustrating search overthinking mechanisms.

Question	Mode	Explanation
Does Disney have an ice princess?	Overthinking	Finds Elsa, then branches into queen/princess and franchise semantics.
Is menthol associated with Thanksgiving?	Spurious association	Cold sensation → winter → holidays, without a true evidential link.
Will Albany, GA reach 100K before Albany, NY?	Step repetition	Population facts are repeated while the required comparison is delayed.
Are months based on the solar cycle?	Overthinking	Direct calendar fact is obscured by historical calendar branches.

Table 10: A* failure mode distribution across all three benchmarks.

Mode	Prop.	Description
Overthinking	91.1%	A* continues past correct evidence, overriding it with spurious conclusions
Premature commitment	8.9%	A* locks onto first branch without sufficient exploration

Failure mode 2: Associative detours. The branching structure can amplify semantically related but logically irrelevant associations. For example, a question about whether menthol is associated with Thanksgiving can lead to a cold-weather branch, then winter, then holiday associations, even though this chain does not support the target relation.

Failure mode 3: Budget dilution. Despite deduplication prompts, A* occasionally generates near-duplicate steps that consume budget without making progress. A representative case shows the model repeatedly restating population information about Albany, Georgia and Albany, New York without completing the comparison required by the question.

7.4 How the Qualitative Patterns Align with Topology

The examples above align with the topology and topic analyses. A* succeeds when there is value in preserving alternatives: non-linear evidence aggregation, hypothesis elimination, multi-hop chaining, and premise scanning. CoT succeeds when the problem admits a compact linear explanation: direct factual comparison, fluent argument evaluation, or single-step world knowledge. A* fails when its own strength becomes excessive—the search keeps expanding after the decisive premise has already been found. This is the concrete mechanism behind the paper’s main thesis: search is neither universally better nor worse; it is useful only when the problem topology rewards exploration.

8 When Search Fails and Amplifies Capability

The qualitative mechanisms above reveal two sides of the same phenomenon. Search helps when branching preserves useful alternatives; it fails when branching creates distractors. We now quantify the dominant failure mode of A* and then test whether search can compensate for limited model capacity when the topology demands multi-hop exploration.

8.1 A* Failure Mode Taxonomy

The main failure modes of the two strategies are mirror images. CoT often fails through premature commitment: it chooses one path too early and cannot recover. A* often fails through overthinking: it continues expanding after the decisive premise has already been found. Thus, the same mechanism that helps A* on branching problems hurts it on linear problems.

We manually classify all A* failure cases into three categories (Table 10).

Overthinking dominates (91.1%): A*’s expansion mechanism is designed for thoroughness, but on single-step problems this becomes pathological—the search continues past the correct answer, accumulating noise. This connects to the

Table 11: CoT vs. A* accuracy across model sizes and benchmarks.

Model	Dataset	CoT	A*	Gain
Qwen-0.5B	HotpotQA	8.40	30.00	+21.6 pp (3.5 $\times$)
Qwen-0.5B	StrategyQA	53.60	53.40	−0.2 pp
Qwen-0.5B	LogicQA	24.00	20.20	−3.8 pp
LLaMA-3.1-8B	HotpotQA	76.80	80.10	+3.3 pp
LLaMA-3.1-8B	StrategyQA	64.52	74.80	+10.3 pp
LLaMA-3.1-8B	LogicQA	45.78	44.85	−0.9 pp

Table 12: Capability $\times$ topology interaction. Cells show A* gain over CoT.

	Compact Logic	Commonsense	Multi-hop
Strong (LLaMA-8B)	−0.9	+10.3	+3.3
Weak (Qwen-0.5B)	−3.8	−0.2	+21.6

overthinking survey (Zhang et al., 2025b) and redundant-state over-exploration (Zhang et al., 2025a); our analysis provides a per-instance quantification of overthinking in search-based reasoning.

8.2 Search as Capability Amplification

We replicate the experiment using Qwen-0.5B (16 $\times$ smaller) to test whether search compensates for limited model capability (Table 11).

Two patterns emerge: (i) **LogicQA degradation is topology-driven**—both model sizes lose, confirming compact deductive reasoning resists search at any capability level; (ii) **HotpotQA amplification is capability-driven**—the weaker model gains far more because A* substitutes the working memory Qwen-0.5B lacks. This connects to rStar-Math (Guan et al., 2025), which showed search helps SLMs in mathematical reasoning. Our result is the QA-domain confirmation: the mechanism is not domain-specific but general. We term this **search-as-capability-amplification**.

8.3 Efficiency Analysis

A* is not more token-efficient in the raw accuracy-per-token sense. Its value is conditional: on branching-topology problems, it reaches accuracy levels that linear CoT or self-consistency do not reach under comparable sampling. Thus, search should be viewed as a targeted allocation of extra computation, not as a default replacement for CoT.

9 Discussion

The preceding sections form a single chain: aggregate results show no universal winner; instance-level outcomes reveal complementarity; topology and topic analyses explain the complementarity; routing experiments show that the explanation is actionable; qualitative and failure analyses identify the mechanism.

The central conclusion of this study is that search and CoT should not be viewed as competing global algorithms. They are different computation policies matched to different input structures. CoT is a low-variance linear policy: efficient when the correct path is direct, brittle when early commitments are wrong. A* is a higher-variance exploration policy: useful when alternatives must be maintained, harmful when exploration creates distractors. Reasoning topology determines which failure mode is more likely.

9.1 The Routing Problem is Topology Recognition

The topology router—which sees zero model outputs—recovers more oracle gap than routers that inspect full reasoning traces. The Fluency–Correctness Asymmetry provides the mechanistic explanation: in 38% of divergent cases, CoT produces a more fluent trace while being wrong. Any router using trace quality as its signal encounters a hard ceiling at

Table 13: Compute efficiency. Token estimates: average node counts ($N_e=3$, 60 tokens/step).

Method / Dataset	Acc.	Tokens	Acc./1K
CoT / StrategyQA	64.52	200	322.6
SC / StrategyQA	71.22	600	118.7
A* / StrategyQA	74.80	972	76.9
CoT / HotpotQA	76.80	200	384.0
SC / HotpotQA	78.21	600	130.4
A* / HotpotQA	80.10	936	85.6

these cases. The topology router bypasses this ceiling by never looking at traces. This reframes adaptive inference: the goal is not merely to verify which trace looks better after generation, but to decide which computational procedure the problem deserves before expensive generation begins.

9.2 Convergence of Topology and Topic Analyses

Our two diagnostic lenses converge on the same conclusion from different angles:

- **Topology view:** High branching coefficient $\rightarrow$ A* advantage (C2, C4 clusters).
- **Topic view:** Multi-hop content categories $\rightarrow$ A* advantage (81.4% of corpus).
- **Shared explanation:** A* wins when problems require systematic exploration of interconnected facts; CoT wins when a single inferential step suffices.

This convergence strengthens confidence: it is not an artefact of one analytical method but a genuine structural property of the question space. The topic model does not replace topology; it stress-tests it. If the topology story were an artefact of hand-crafted features, we would not expect independent lexical-topic categories to separate into the same multi-hop versus single-step division.

9.3 The 73.9% Residual Gap

The topology router recovers 26.1% of the oracle gap, leaving 73.9% unreachable. Closing this residual requires discriminating within a topology class—distinguishing two multi-hop questions where one aligns with the model’s parametric knowledge and one does not. This likely requires **lightweight problem–model compatibility scoring**: a measure of how well the base model’s parameters support the retrieval operations the problem demands. The remaining gap suggests that topology is necessary but not sufficient. A router also needs to know whether the model has the knowledge required to execute the needed search. Two questions can have the same topology while differing in parametric familiarity, retrieval difficulty, or susceptibility to hallucination.

9.4 Broader Implications

Our findings have direct practical implications for LLM deployment. For applications where questions can be pre-analysed (search engines, QA systems, educational tutoring), topology-based pre-routing reduces compute cost while maintaining performance. For SLM deployment at the edge, search-as-capability-amplification provides a path to competitive performance without larger models.

9.5 Limitations

1. **Confidence miscalibration.** Self-reported confidence is a ranking signal, not a calibrated probability. Future work: calibrated uncertainty-aware search.
2. **Single model family.** Results are based on LLaMA 3.1-8B and Qwen-0.5B. Extension to reasoning-native models (o1-mini, DeepSeek-R1) is needed.

3. **No fine-tuning.** All comparisons use prompting. Jointly training routing and search policies may further improve performance.
4. **Symbolic features.** Our 13 features are hand-crafted. Learning topology embeddings from question encodings may capture richer signals.
5. **English-only.** Topology features based on dependency parsing may not transfer directly to other languages.
6. **Search-policy dependence.** Our conclusions are based on one budgeted A* implementation with a specific critic, branching factor, prompt, and stopping rule. Other search policies may shift the boundary between helpful and harmful search; the goal of this paper is to diagnose the boundary, not to claim that this implementation is the final optimal search procedure.
7. **Topic modelling is descriptive.** Topic modelling is used as a descriptive validation tool, not as a causal explanation. The NMF categories help show that topology-sensitive patterns cut across datasets, but the precise topic boundaries depend on preprocessing and the choice of k .
8. **Topology features are proxies.** Our topology features are observable proxies for reasoning structure. They do not directly measure the latent causal structure of the reasoning process. More principled topology representations may be learned from proof graphs, retrieval graphs, or counterfactual perturbations.

10 Conclusion

This paper studied when guided search should be preferred to linear chain-of-thought reasoning. Our results show that the answer cannot be obtained from aggregate benchmark accuracy alone. A* improves over CoT on some datasets and not on others, but the more important pattern appears at the instance level: the two strategies solve partially different subsets of problems.

We argued that this complementarity is explained by reasoning topology. Search helps when a problem requires preserving alternatives, scanning multiple premises, chaining interdependent facts, or resolving competing entity paths. CoT is preferable when the answer follows from a compact linear inference. This claim is supported by topology clustering, cross-dataset topic modelling, topology-aware routing, qualitative analysis, and the failure taxonomy.

The practical implication is that adaptive inference should be framed as problem-structure recognition. Post-hoc trace inspection is limited because fluent traces can be wrong and search traces can appear less coherent despite tracking the decisive evidence. A router that observes only pre-generation topology features already recovers more oracle gap than output-based routers, suggesting that the decision of whether to search should often be made before generation begins.

Overall, search is not a universally stronger replacement for CoT. It is a higher-cost, higher-variance computation strategy whose value depends on the structure of the problem and the capability of the underlying model. The next step is to close the remaining oracle gap by modelling not only problem topology, but also problem-model compatibility: whether the model has the knowledge and representational capacity needed to exploit the search space it is given.

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A Additional Formal Details and Qualitative Examples

This appendix provides additional formal details, routing definitions, qualitative examples, and future-work ideas supporting the main analysis.

A.1 Complementarity as the Starting Point

A useful way to frame the problem is as multi-strategy inference: CoT and A* are complementary reasoning procedures, and a practical system should exploit this complementarity through routing. The core observation is that CoT commits to one path, whereas A* searches a tree of possible reasoning steps. The main paper builds on this observation by asking what structural properties of the input explain the complementarity.

A.2 Additional Routing Definitions

We consider three auxiliary routing mechanisms. Semantic entropy estimates uncertainty from multiple CoT samples and routes high-entropy examples to A*. The rule-based random forest extracts hand-crafted problem features such as question length, number of logical connectives, negation, and question type. The LLM-as-critic router invokes DeepSeek-R1-Distill-Qwen-14B only when CoT and A* disagree; the critic receives both traces and emits a structured decision about which trace better supports its answer. The TMLR topology router should be read as a more principled version of the feature-based route: it replaces ad hoc surface features with an explicit reasoning-topology feature set.

A.3 Interpretation of Critic Limitations

The qualitative analysis suggests that LLM-as-critic routing improves only modestly because A* traces are often non-linear and step-wise, while CoT traces are fluent and narratively coherent. A language-model critic may therefore prefer the CoT trace even when A* reaches the correct answer through less fluent but more rigorous branching. The main analysis measures this phenomenon through the Fluency–Correctness Asymmetry: in a substantial fraction of divergent cases, the more fluent trace is wrong.

A.4 Additional Qualitative Examples

Spatial reasoning in LogicQA. A CoT trace can accumulate errors when processing spatial constraints sequentially. A* can explore multiple spatial assignments and find one that simultaneously satisfies the constraints. This is another instance of the hypothesis-preservation benefit of search.

Average number of peas and commas for a billion. Additional examples show cases where CoT succeeds because the answer depends on fluent world knowledge or direct numerical comparison. Such cases serve as CoT-dominant examples where additional branching is unnecessary.

Trace normalization as future work. A useful future direction is trace normalization: rewriting A*’s branched reasoning into a coherent linear narrative before presenting it to an LLM critic or human judge. This may reduce the bias against search traces in post-generation routing.

B Alternative Conservative Heuristic Construction

This appendix records an auxiliary depth–coverage heuristic construction. It is not the primary heuristic used in the main balanced experiments, but it provides a constructive way to think about conservative search guidance when minimum-hop assumptions and explicit coverage terms are available.

Let $c(n, n') = 1 + (1 - c_t)$ be the transition cost for a reasoning step with self-reported confidence $c_t \in [c_{\min_conf}, c_{\max_conf}]$, so the minimum single-step cost is $c_{\min} = 1 + (1 - c_{\max_conf})$. The auxiliary heuristic uses

$$h(n) = h_{\text{depth}}(n) + h_{\text{coverage}}(n), \quad (8)$$

with

$$h_{\text{depth}}(n) = \omega_d \max(0, M - d_n), \quad h_{\text{coverage}}(n) = \omega_c(1 - E(n)), \quad (9)$$

where M is a minimum hop constraint, d_n is the node depth, and $E(n)$ is the fraction of target entities explicitly addressed in the partial trace. Under constraints such as $\omega_d M + \omega_c \leq M c_{\min}$ and $\omega_c \leq c_{\min}$, the heuristic can be bounded below the minimum remaining cost in the root, intermediate-depth, and deep-node cases. This construction is useful as a formal sanity check, while the main experiments use the critic-based heuristic because it applies uniformly across all three datasets.

C Alternative Formal Proof Sketch for the Depth–Coverage Heuristic

This proof sketch applies to the auxiliary heuristic in Appendix B, not to the main critic-based heuristic.

For A* search to guarantee an optimal solution in the classical sense, the heuristic must never overestimate the true minimum remaining cost. Consider the heuristic $h(n) = \omega_d \max(0, M - d_n) + \omega_c(1 - E(n))$. Let $c_{\min} = 1 + (1 - c_{\max_conf})$ be the minimum cost of a valid reasoning step. If $\omega_d M + \omega_c \leq M c_{\min}$ and $\omega_c \leq c_{\min}$, then:

1. **Root node.** At $d_n = 0$, at least M steps are required, so $h^*(n) \geq M c_{\min}$. The maximum heuristic value is $\omega_d M + \omega_c \leq M c_{\min}$.
2. **Intermediate nodes.** For $0 < d_n < M$, at least $M - d_n$ steps remain, so $h^*(n) \geq (M - d_n) c_{\min}$. The depth term decreases with d_n , and the coverage penalty is bounded by ω_c .
3. **Deep non-goal nodes.** For $d_n \geq M$, the depth term is zero, but a non-goal node requires at least one more transition. Hence $h^*(n) \geq c_{\min}$ and $h(n) \leq \omega_c \leq c_{\min}$.

The purpose of including this proof is not to claim that the final critic-based A* implementation inherits the same exact guarantee. Rather, it shows how a search heuristic over reasoning traces can be made conservative when the cost structure and minimum-hop assumptions are explicitly controlled.

D Qualitative Examples

D.1 A* Success Cases

Non-linear evidence aggregation (StrategyQA). “*Is shrimp scampi definitely free of plastic?*” (Gold: NO). CoT analyses preparation steps sequentially and concludes YES. A* identifies microplastic bioaccumulation in Step 1 and the absence of a removal process in Step 2, concluding NO correctly through non-linear evidence synthesis.

Hypothesis elimination (LogicQA). Modus tollens identification. CoT maps to the modus ponens option due to surface similarity. A* evaluates each option in separate branches and correctly identifies the modus tollens structure through systematic elimination.

Multi-hop chaining (HotpotQA). “*What government position was held by the woman who portrayed Corliss Archer?*” (Gold: Chief of Protocol). CoT identifies Shirley Temple but confuses government roles. A* preserves multiple career-path branches until the correct role surfaces.

D.2 CoT Success Cases

Direct factual recall (StrategyQA). “*Is a Boeing 737 cost covered by Wonder Woman (2017) box office receipts?*” (Gold: YES). CoT retrieves magnitudes directly. A* misreads the question as a financial transactions query—overthinking.

Fluent argumentation (LogicQA). Property-transfer argument. CoT maintains the global argumentative frame. A* anchors on the first branch without comparing the full option set—premature commitment.

Table 14: Representative A* failure cases.

Question	Mode	Explanation
Does Disney have an ice princess?	Overthinking	Elsa found correctly, then queen/princess debate
Is Menthol associated with Thanksgiving?	Spurious assoc.	Cold weather → winter → holiday chain
Will Albany, GA reach 100K before Albany, NY?	Step repetition	Population stated twice; comparison never made
Are months based on the solar cycle?	Overthinking	Direct recall; A* branches into calendar history

D.3 Representative Failure Cases

E Human Evaluation Details

Table 15: Human annotation on 100 sampled disagreement cases per dataset. Cov. = coverage (1–3); Rel. = relevance (1–3). Cohen’s κ ranges 0.59–0.68.

Dataset	Group	CoT Cov.	CoT Rel.	A* Cov.	A* Rel.
StrategyQA	Both wrong ($n=11$)	1.00	2.82	1.09	1.55
	CoT right ($n=51$)	3.00	2.57	1.20	2.16
	A* right ($n=38$)	1.63	2.47	2.71	2.00
LogicQA	Both wrong ($n=38$)	1.79	2.76	1.89	2.53
	CoT right ($n=36$)	3.00	2.81	1.86	2.47
	A* right ($n=26$)	1.88	2.81	2.77	2.42

Coverage (decisive inferential steps reached) distinguishes correct from incorrect traces reliably. Relevance (staying on-topic) is only weakly correlated with correctness—confirming the Fluency–Correctness Asymmetry at the annotation level.

F Prompt Templates

CoT prompt.

Question: {question}
Context: {context}
Let’s think step by step.

A* expansion prompt.

You are solving a reasoning problem step by step.
Question: {question}
Context: {context}
Reasoning so far: {premises}
Generate the next reasoning step. Consider:
(a) supporting evidence, (b) counterarguments,
(c) factual lookups.
Provide exactly one step.

Critic scoring prompt.

Rate the following partial reasoning on a scale

of 0.0 to 1.0:
Question: {question}
Context: {context}
Reasoning: {reasoning_trace}
Score based on: (1) factual consistency with
context, (2) reasoning completeness toward
answering the question.
Output only a number between 0.0 and 1.0.

Self-Consistency prompt. Same as CoT prompt, with temperature $\tau = 0.9$ and 5 independent samples. The majority-voted answer is selected.

G Code and Data Availability

All code, experimental configurations, and result files will be released upon acceptance at [https://github.com/\[anonymised\]](https://github.com/[anonymised]). The datasets are publicly available from their original sources. All experiments were conducted on a single NVIDIA A100 GPU with Ollama v0.5 serving LLaMA 3.1-8B and DeepSeek-R1-Distill-Qwen-14B.

H TMLR Reproducibility Checklist

- ✓ All datasets are publicly available.
- ✓ Model weights are publicly available (LLaMA 3.1-8B via Meta, Qwen-0.5B via Alibaba, DeepSeek-R1-Distill-Qwen-14B via DeepSeek).
- ✓ Hyperparameters reported (§3.5).
- ✓ Random seeds fixed (42 throughout).
- ✓ Prompt templates provided (Appendix F).
- ✓ Confidence intervals reported (Table 2).
- ✓ Statistical significance tests reported (McNemar’s, Table 2).
- ✓ Code to be released (Appendix G).